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Sea level controls terrigenous sediment provenance evolution in the Okinawa Trough since the Last Glacial Maximum

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The Okinawa Trough (OT), a typical back-arc basin in the western Pacific, preserves continuous sedimentary records since the Quaternary, making it a critical area for investigating sea-level changes, source-to-sink processes, and paleoclimate evolution. However, the controlling mechanisms of terrestrial input to the trough since the Last Glacial Maximum (LGM) remain insufficiently understood. This study conducted major element geochemical analysis on core C1624 from the southern part of the middle OT, combined with AMS¹⁴C dating, grain size, and mineralogical data, to reconstruct the evolution and dominant controls of terrestrial input since the LGM. Principal component analysis (PCA) indicates that the sediments in the study area are primarily composed of a mixture of two end-members: terrigenous detritus and marine biogenic carbonate. The ratios of TFe₂O₃/Al₂O₃ and CaO/Al₂O₃ were adopted as reliable proxies for terrestrial input and biogenic contribution, respectively. The results reveal a three-stage response of terrestrial input to sea-level changes since the LGM: Stage 1 (25.4–11.6 ka BP), during low sea-level stands, the paleo-Changjiang river system extended to the outer shelf, supplying the dominant sediment source; Stage 2 (11.6–8.7 ka BP), characterized by rapid sea-level rise, witnessed a transition in provenance from the Changjiang to Taiwan-derived materials, accompanied by a sharp decrease in terrestrial input; Stage 3 (8.7–0 ka BP), with the establishment of modern circulation patterns, Taiwan-sourced materials became predominant. This study demonstrates that sea-level changes on glacial-interglacial scales are the fundamental factor reshaping the depositional pattern and driving provenance transitions in the middle OT. These findings provide a new case for understanding the response of marginal sea sediment source-to-sink processes to global climate change.

KEYWORDS

Last Glacial Maximum, Okinawa Trough, sea level, sediment provenance, terrestrial input

1 Introduction

The Okinawa Trough (OT), a typical back-arc basin in the western Pacific, is a narrow arc-shaped basin situated between the East China Sea (ECS) continental shelf and the Ryukyu Island Arc. It serves as an ideal archive for investigating land-sea interactions, source-to-sink processes, and paleoclimate evolution due to its continuous deposition of high-resolution paleoenvironmental records (Jian et al., 2000; Li et al., 2001; Meng et al., 2009; Xu et al., 2018; Zheng et al., 2014). The sedimentary environment in this region is complex, with diverse material sources influenced by a combination of factors including sea-level changes (Diekmann et al., 2008; Li et al., 2018; Wang and Sun, 1994), the East Asian monsoon (An et al., 2000; Cheng et al., 2016; Kao et al., 2006a), and the pathway and intensity of the Kuroshio Current (KC) (Jian et al., 1998; Kao et al., 2006b; Li et al., 2019; Ujiie and Ujiie, 1999; Ujiie et al., 2003). These sedimentary archives not only form the foundation for reconstructing regional paleoclimate and paleoceanographic evolution but also provide a critical example for understanding the transport of materials and energy from the continent to the ocean during global glacial-interglacial cycles. Consequently, the study of Quaternary sedimentation in the OT has emerged as a forefront research area in international paleoceanography (Chen et al., 2025; Kao et al., 2008; Wang et al., 2026, 2025).

Over the past two decades, significant progress has been made in understanding sediment provenance in the OT, facilitated by the widespread application of high-precision dating techniques and multi-proxy analytical methods. Geochemical tracers, such as major and trace elements (Jiang et al., 2011; Li et al., 2024; Xu et al., 2025), rare earth elements (Dou et al., 2010a; Xu et al., 2012, 2019), and Sr-Nd isotopes (Dou et al., 2012; Hu et al., 2020; Li et al., 2015), along with mineralogical analyses including clay minerals (Diekmann et al., 2008; Dou et al., 2010b; Wang et al., 2025) and magnetic minerals (Liu et al., 2007a), as well as biological components (Zhang et al., 2010) have become mainstream approaches for deciphering sediment sources in this region.

Studies indicate that the potential material sources for the OT are complex and dynamic, primarily including (Diekmann et al., 2008; Dou et al., 2010a; Jiang et al., 2010, 2019; Katayama and Watanabe, 2003; Wang et al., 2024; Zou et al., 2022): (1) the Changjiang and Huanghe from Chinese mainland; (2) sediments derived from the Taiwan orogen; (3) biogenic carbonate materials; and (4) inputs from volcanic eruptions and submarine hydrothermal activities originating from the Ryukyu Island Arc and within the trough itself.

However, the response of these different source materials to sea-level changes since the Last Glacial Maximum (LGM) remains poorly constrained. To address this scientific question, this study focuses on sediment core C1624 collected from the southern part of the middle OT (Figure 1). Through systematic geochemical analysis of the sediments, utilizing major elements combined with published data on grain size, AMS¹⁴C dating, and mineralogical indicators (Wang et al., 2025), this study aims to select appropriate proxies for terrestrial input and investigate its response to sea-level changes in the middle OT since the LGM.

2 Regional setting

The OT is located on the outer edge of the ECS continental shelf, west of the Ryukyu Islands. It is a NE-SW trending, arc-shaped back-arc basin (Figure 1), extending approximately 1200 km from Kyushu, Japan, in the north to Taiwan, China, in the south, with a width of 140–200 km. The trough topography is generally narrower in the middle part and wider in the northern and southern sections, with water depth gradually increasing from north to south (Li et al., 2016a; Liu et al., 2007a). As a typical active back-arc basin in the western Pacific, its formation resulted from the northwestward subduction of the Philippine Sea Plate beneath the Eurasian Plate since the late Miocene (Letouzey and Kimura, 1986; Sibuet et al., 1987). It is characterized by generally higher heat flow values than the surrounding areas and widespread development of submarine volcanoes, hydrothermal vents, and cold seep systems (Ishibashi et al., 2015; Zou et al., 2022). These geological features indicate that the OT is an active tectonic unit (Huh et al., 2006), however, its sedimentation since the late Quaternary has been predominantly controlled by sea-level changes and climate rather than tectonic activity.

The geological evolution of the ECS shelf since the last glacial period has been governed by global sea-level fluctuations, resulting in a complete shelf sedimentary sequence (Saito et al., 1998). Li et al. (2014) indicate that since sea level dropped to approximately -135 m during the LGM, the ECS has experienced four rapid transgressive phases and three slow sea-level rise phases. Correspondingly, its sedimentary systems sequentially developed into: a lowstand systems tract (featuring paleo-coastal zones, slope progradational wedges, and turbidite deposits), a transgressive systems tract (forming tidal sand ridges and transgressive boundary layers), and a highstand systems tract (establishing modern mud deposits controlled by contemporary circulation patterns). Sea-level fluctuations exert a significant controlling influence on material input and sedimentary environment evolution in the OT region (Meng et al., 2009; Zhao et al., 2017).

The marine environment of the OT is primarily governed by two major current systems: the KC-Tsushima Warm Current–Yellow Sea Warm Current system in the east, and the Yellow Sea Coastal Current–ZheMin Coastal Current system in the west (Figure 1). Among these, the KC, the strongest western boundary current in the western Pacific (Hu et al., 2015; Lee et al., 2013), flows northeastward along the trough. It is capable of transporting sediments from Taiwan to the southern and even middle parts of the trough (Dou et al., 2016; Li et al., 2015; Xu et al., 2012), profoundly influencing the regional hydrographic environment, material transport, and depositional processes (Ding et al., 2022; Li et al., 2020; Wang et al., 2024; Xiang et al., 2003). Characterized by high temperature, high salinity, large volume transport, and relatively low nutrient levels (Hu et al., 2015), variations in its intensity and axis position not only directly affect marine primary productivity but also act as a “hydrodynamic barrier,” restricting the lateral transport of terrestrial materials from the ECS shelf into the trough (Li et al., 2019; Zheng et al., 2016). Furthermore, studies have shown that the evolution of the KC is closely linked to climate systems such as the El Niño–Southern Oscillation (ENSO) and the

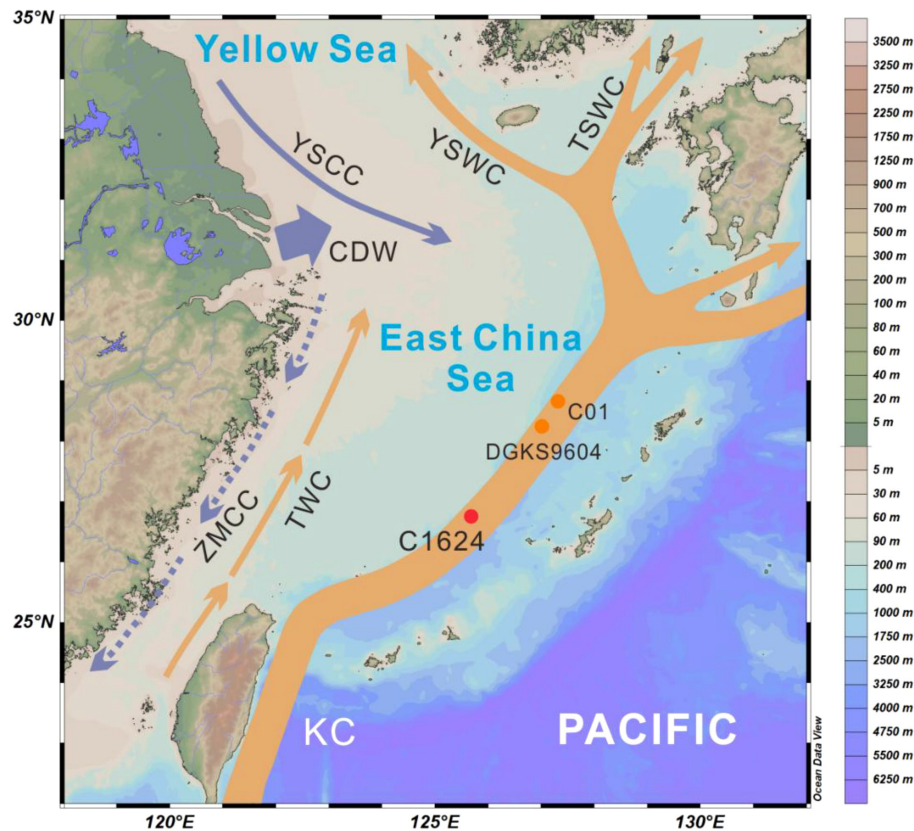


FIGURE 1

Map showing the location of the core C1624 (red circle) and other core locations mentioned in the main text (orange circle). The oceanic circulation, modified from Lian et al. (2016) and Lie and Cho (2016). YSCC, the Yellow Sea Coastal Current; CDW, the Changjiang Diluted Water; ZMCC, the Zhe-Min Coastal Current; TWC, the Taiwan Warm Current; KC, the Kuroshio Current; TSWC, the Tsushima Warm Current; YSWC, the Yellow Sea Warm Current.

East Asian monsoon (Hu et al., 2015, 2020; Kubota et al., 2017; Zheng et al., 2016).

In addition to the KC, the East Asian monsoon (Sun et al., 2011; Zhang et al., 2024; Zheng et al., 2014) is another key climatic driver controlling sediment transport in this region. Modern hydrodynamic observations further reveal that the winter coastal current transport and summer cold-water mass entrapment, co-modulated by the East Asian monsoon and the KC, are key mechanisms in the suspended sediment “source-to-sink” process (Li et al., 2016b). Moreover, the East Asian monsoon can influence sea surface temperature in the OT by modulating precipitation (Xu et al., 2021).

3 Materials and methods

3.1 Core C1624

In 2016, core C1624 was drilled from the western slope of the middle OT (125.685°E, 26.755°N) by the Qingdao Institute of Marine Geology (Figure 1, red circle). The core was collected at a water depth of 1000 m and had a total length of 570 cm. The

sediments of core C1624 are primarily composed of clayey silt (Figure 2A). A distinct sand layer is observed at the bottom of the core, representing the coarsest grain-size interval within the entire core (Figures 2A, B). Upcore, the sand content gradually decreases while the clay content increases, leading to an overall fining upward trend in grain size (Figures 2A, B). However, significant fluctuations in sand and clay contents occur at approximately 170 cm depth, marked by an increase in sand and a decrease in clay (Figure 2A). The mean grain size (M_z) of the core sediments ranges from 5.3 to 29.5 μm , with an average of 10 μm .

Based on the lithological characteristics of core C1624, clean and undamaged monospecific planktonic foraminifera *Neoglobobulimina dutertrei* were selected at various depths (specific depths indicated in Figure 2C). These samples were sent to Beta Analytic Incorporated in the USA for AMS¹⁴C dating. The conventional ¹⁴C ages were calibrated to calendar ages, with the bottom age of core C1624 subsequently estimated to be 25.3 ka based on a linear age-depth model (Wang et al., 2025). Sedimentation rates vary between 8.2 and 370.4 cm/ka. For most of the core depth, the sedimentation rate remains below 25 cm/ka (Figure 2D). An anomalously high sedimentation rate phase occurred between 12.3 and 13.7 ka, during which rates sharply increased to 114.1–370.4 cm/ka (Figure 2D).

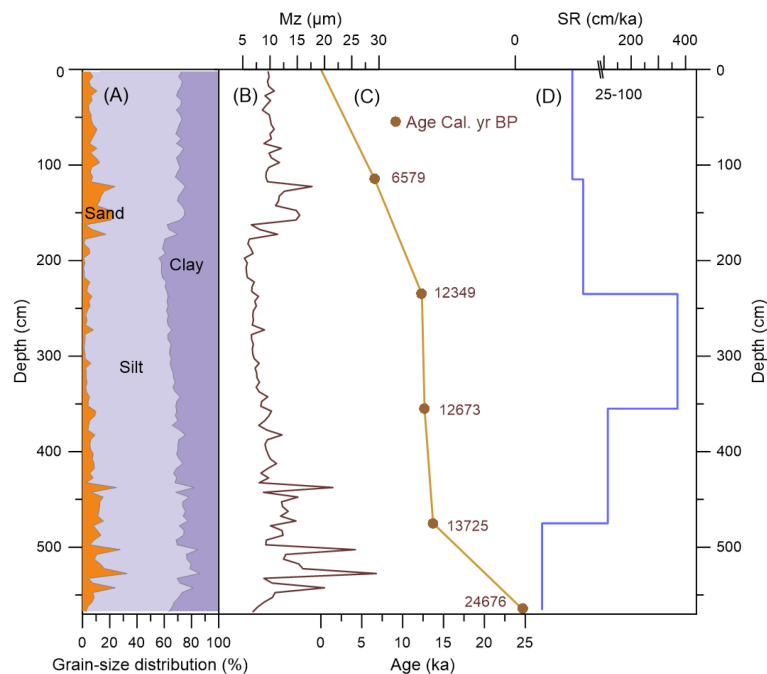


FIGURE 2

Downcore variations in (A) sand, silt, and clay contents (%), (B) mean grain size (Mz; μm), (C) age model with ^{14}C dating points, and (D) sedimentation rates (cm/ka) for core C1624 [date from Wang et al. (2025)].

3.2 Measurements of elemental concentrations

The elemental analysis of sediments was completed at the Qingdao Institute of Marine Geology, China Geological Survey. A total of 56 samples were collected at 10 cm intervals for subsequent analysis. Firstly, sediment samples were dried in an oven at 60 °C for 24 hours and then ground into 200 mesh powders using an agate mortar. Approximately 4 g of the prepared sample was weighed, placed into a mold, and pressed into pellets for analysis. The analysis was performed using a PANalytical Axios X-ray fluorescence (XRF) spectrometer from the Netherlands. The operating conditions were set at a high power of 4 kW, with a maximum excitation voltage of 60 kV, a maximum current of 120 mA, and high transmittance, utilizing an SST ultra-sharp long-life ceramic end-window (75 μm) Rhodium target X-ray tube. The SuperQ software was used to scan the vicinity of elemental spectral line peaks, carefully selecting background positions and identifying interfering elements. Quality control during the analytical process was maintained using national standard reference materials and replicate samples, with the relative standard deviation of the analytical results being less than 5%.

4 Results

SiO_2 is the dominant component, ranging from 42.20% to 54.40% (average 49.05%) and exhibiting distinct stage-wise variations down-core (Figure 3A). The Al_2O_3 content varies between 13.08% and 16.61%, with an average of 15.11%. Its

down-core profile reveals three distinct stages (Figure 3B): from the base to ~230 cm, values fluctuate within a narrow range with a slight upward-increasing trend; between ~230 and 160 cm, values change markedly, showing a brief increase followed by a sharp decrease; from ~160 cm to the top, Al_2O_3 content remains at the core's lowest overall levels and increases slowly upward.

K_2O content ranges from 2.96% to 4.18% (average 3.68%). MgO content is relatively low, varying from 2.40% to 3.08% (average 2.83%). TiO_2 content is extremely stable and low, concentrated within a narrow range of 0.60% to 0.74% (average 0.67%), representing a typical stable terrigenous component. Total Fe_2O_3 (TFe_2O_3) content fluctuates between 4.63% and 6.57%, with an average of 5.83%. The overall down-core trends of K_2O , MgO , TiO_2 , and TFe_2O_3 are remarkably consistent with those of Al_2O_3 content described above (Figures 3C–F).

MnO maintains the lowest concentrations among major elements, consistently below 0.1% (average 0.07%), and generally decreases slowly upward (Figure 3G). CaO content exhibits a large variation, ranging from 5.13% to 14.1% (average 7.74%). Its overall down-core trend is opposite to those of Al_2O_3 , K_2O , MgO , TiO_2 , and TFe_2O_3 . CaO also displays three clear stages: persistently low values with minor fluctuations from the base to ~230 cm; a slight decrease following a rapid increase between ~230 and 160 cm; and the core's highest overall levels from ~160 cm upward, with a slow decreasing trend toward the top (Figure 3H). Na_2O content varies between 2.96% and 4.18% (average 3.68%), showing an overall increasing, albeit fluctuating, trend upward (Figure 3I).

In summary, the down-core variation patterns differ among components, indicating distinct depositional controls. For instance, CaO shows a seesaw relationship with components like Al_2O_3 and TFe_2O_3 at certain depths. More detailed geochemical data are

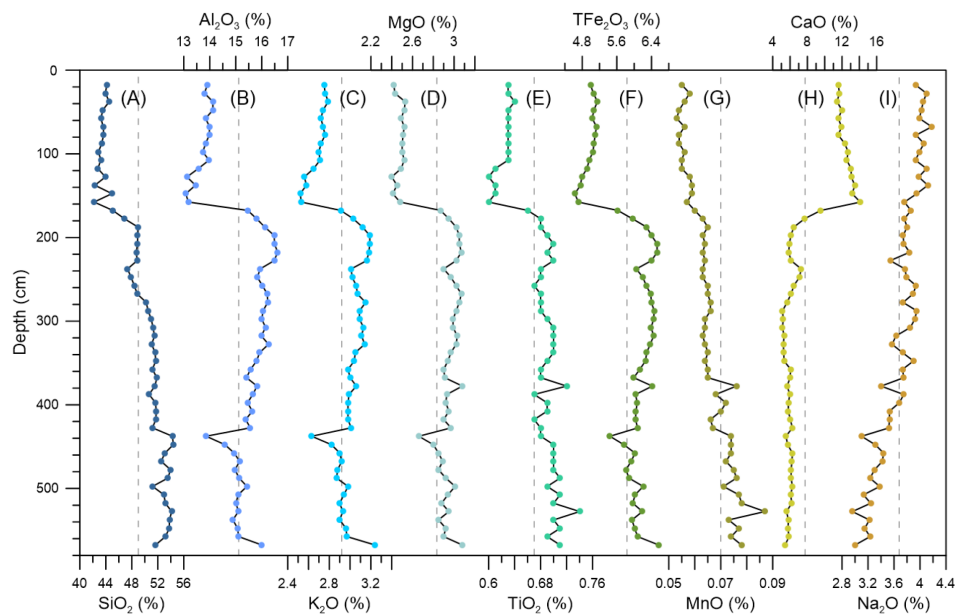


FIGURE 3

Variation of SiO_2 contents (A), Al_2O_3 contents (B), K_2O contents (C), MgO contents (D), TiO_2 contents (E), TFe_2O_3 contents (F), MnO contents (G), CaO contents (H), and Na_2O contents (I) against the depth of core C1624. The dashed lines represent their respective mean values.

provided in [Supplementary Table 1](#). The combined characteristics of these geochemical parameters systematically record the history of sediment provenance shifts and depositional environment evolution experienced by core C1624.

5 Discussions

5.1 Sediment provenance

The major element composition of marine sediments serves as a key geochemical indicator for interpreting their provenance and depositional processes ([Jung et al., 2023](#); [Liu et al., 2017](#)). Sediments in the OT are typically a mixture of terrestrial detrital input, marine biogenic components, and potential volcanic/hydrothermal contributions ([Qin et al., 1996](#)). Accurate identification and quantification of these end-members are fundamental for reconstructing paleoenvironments and depositional patterns. Principal component analysis (PCA), an effective multivariate statistical method, can reduce the dimensionality of complex geochemical datasets ([Nesrstová et al., 2025](#)), thereby identifying the main common factors controlling element distributions and providing a reliable basis for tracing material sources.

To reveal the characteristics of sediment sources in the southern middle OT, we performed PCA on the contents of SiO_2 , Al_2O_3 , CaO , MgO , K_2O , Na_2O , MnO , TiO_2 , and TFe_2O_3 in core samples using SPSS 27.0 software ($\text{KMO} = 0.789$, $p < 0.001$). The analysis extracted two principal components (PC1 and PC2), which together accounted for 96.1% of the total variance (PC1: 78.6%; PC2: 17.5%) ([Figure 4A](#)).

The loading plot reveals that PC1 accounts for the dominant portion. Its positive loading is strongly associated with SiO_2 , Al_2O_3 ,

MgO , K_2O , MnO , TiO_2 , and TFe_2O_3 , while CaO shows a significant negative loading ([Figure 4A](#)). This pattern clearly indicates that PC1 represents the mixing relationship between terrestrial detrital input (positive end) and marine biogenic carbonate input (negative end) ([Jiang et al., 2011](#)), which is the predominant process controlling sediment composition in the study area. PC2 is primarily associated with the positive loading of Na_2O ([Figure 4A](#)). The score plot visually confirms this end-member mixing relationship. The sample points show a good continuous distribution along the PC1 axis, and their color (representing CaO content) systematically transitions from high values on the left to low values on the right ([Figure 4B](#)), confirming a strong negative correlation between carbonate content and PC1 scores. This implies that as the proportion of terrigenous detrital material increases (higher PC1 score), the carbonate component may be diluted.

Core C1624 is located on the western slope of the trough, whereas volcanic materials in the OT are mainly enriched on the eastern slope ([Xu et al., 2025](#)). Furthermore, no distinct tephra layers were identified in core C1624 ([Figure 2](#)). Additionally, elements typically considered sensitive indicators of hydrothermal activity, such as TFe_2O_3 , MgO , and MnO ([Li et al., 2021](#); [Yang et al., 2015](#)), are highly aggregated at the positive end of PC1 in this PCA result, showing strong positive correlations with stable terrigenous elements like Al_2O_3 , K_2O , and TiO_2 . Moreover, there is a lack of significant factors independent of PC1 and PC2 that are dominated by typical hydrothermal element associations (e.g., Fe-Mg-Mn-rich). While dissolved or colloidal hydrothermal iron could potentially adsorb onto fine-grained sediments ([Li et al., 2021](#)), its influence is likely very limited. Therefore, it is suggested that iron in sediments is primarily derived from terrestrial rock weathering and erosion, being transported and deposited in detrital form, with its distribution controlled by a unified terrestrial input process ([Plewa et al., 2012](#)), rather than by independent hydrothermal activity.

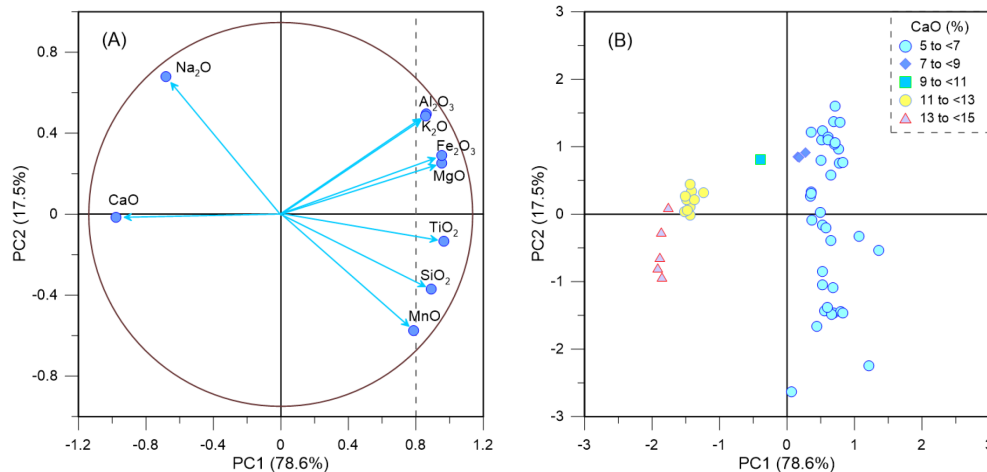


FIGURE 4

Principal component analysis (PCA) is expressed by the loading plots (A) and score plots (B). The dashed vertical line at PC1 = 0.8 serves as a threshold for identifying variables with high loadings on PC1.

Although authigenic carbonate formed via microbial-mediated early diagenetic processes (e.g., anaerobic oxidation of methane) is a recognized carbonate factory in marine sediments (Dai et al., 2025), its influence on the bulk sediment CaO content in the study area is considered minor. A previous geochemical study of core CSHC-4 from the OT has quantified that such methane-seep-related authigenic carbonates contribute only a small fraction (averaging ~3.7%) to the total carbonate pool (Li et al., 2024). Therefore, the PCA results effectively support the conclusion that sediment provenance in the study area is primarily controlled by terrigenous detritus and biogenic carbonate.

5.2 Influence of terrigenous dilution

Sediment particle characteristics significantly influence elemental concentrations (Yang et al., 2003). To accurately assess the terrestrial input signal and eliminate the potential interference of sediment grain size variations on geochemical elemental indicators, we conducted a systematic correlation analysis between key terrigenous (Fe, Al) and biogenic (Ca) elemental indicators and sediment grain-size parameters (Figure 5).

Firstly, there is no significant correlation between TFe_2O_3 content and the Mz, which reflects the overall grain-size level of the sediment ($r = 0.32$, $p < 0.05$; Figure 5A). This indicates that fluctuations in TFe_2O_3 content are not primarily driven by simple overall coarsening or fining of the sediment. However, TFe_2O_3 shows a significant positive correlation with the clay fraction content ($Y = 4.43 \times X + 5.15$, $r = 0.44$, $p < 0.01$; Figure 5B). This aligns with previous studies in marginal seas, where fine-grained particles, especially clay minerals, exhibit a strong adsorption and carrying capacity for iron (hydr)oxides due to their larger specific surface area (Gu et al., 2025; Kong et al., 2024). Aluminum (Al) is typically regarded as a classic tracer for terrigenous clay minerals. It remains stable during depositional processes and is largely

unaffected by marine biological processes and diagenesis (Dymond et al., 1997; Feng et al., 2021; Murray and Leinen, 1996). In recent years, the method of elemental normalization using Al to eliminate physical interferences such as grain size has been widely applied (Sun et al., 2008). As shown in Figure 5C, TFe_2O_3 and Al_2O_3 contents exhibit an extremely strong positive correlation ($Y = 1.66 \times X + 5.44$, $r = 0.96$, $p < 0.01$). This strongly confirms that, in the study area, both are jointly and predominantly controlled by the input of terrigenous clay minerals, rather than being significantly affected by later diagenetic alteration.

For calcium, an indicator element for biogenic carbonate (Jiang et al., 2011), the content of its oxide (CaO) shows no clear correlation with the Mz (Figure 5D). Further analysis indicates that CaO has no significant association with any of the sand, silt, or clay fractions (Figure 5E), demonstrating that its distribution is not controlled by sediment grain size. However, the PCA results show that CaO is strongly negatively correlated with the PC1 factor, which represents the dominance of terrigenous detritus (Figure 4). This relationship is directly manifested in elemental contents as a high negative correlation between CaO and Al_2O_3 ($Y = -0.29 \times X + 17.349$, $r = -0.85$, $p < 0.01$; Figure 5F).

As elemental ratios are insensitive to the closure effect, they are often more indicative of material input than individual elemental concentrations (Weltje and Tjallingii, 2008). Based on the above comprehensive analysis, this study adopts the $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratio to indicate the relative enrichment degree of terrestrial material input (Jung et al., 2023), thereby excluding the simple grain-size enrichment effect. The $\text{CaO}/\text{Al}_2\text{O}_3$ ratio is used to more reliably indicate changes in the contribution of biogenic carbonate relative to the total terrigenous detritus. These two ratio indicators effectively circumvent the influence of grain size on elemental concentrations (Xu et al., 2025), thus providing a reliable geochemical basis for subsequent discussion on the evolution of terrestrial input under the context of sea-level changes.

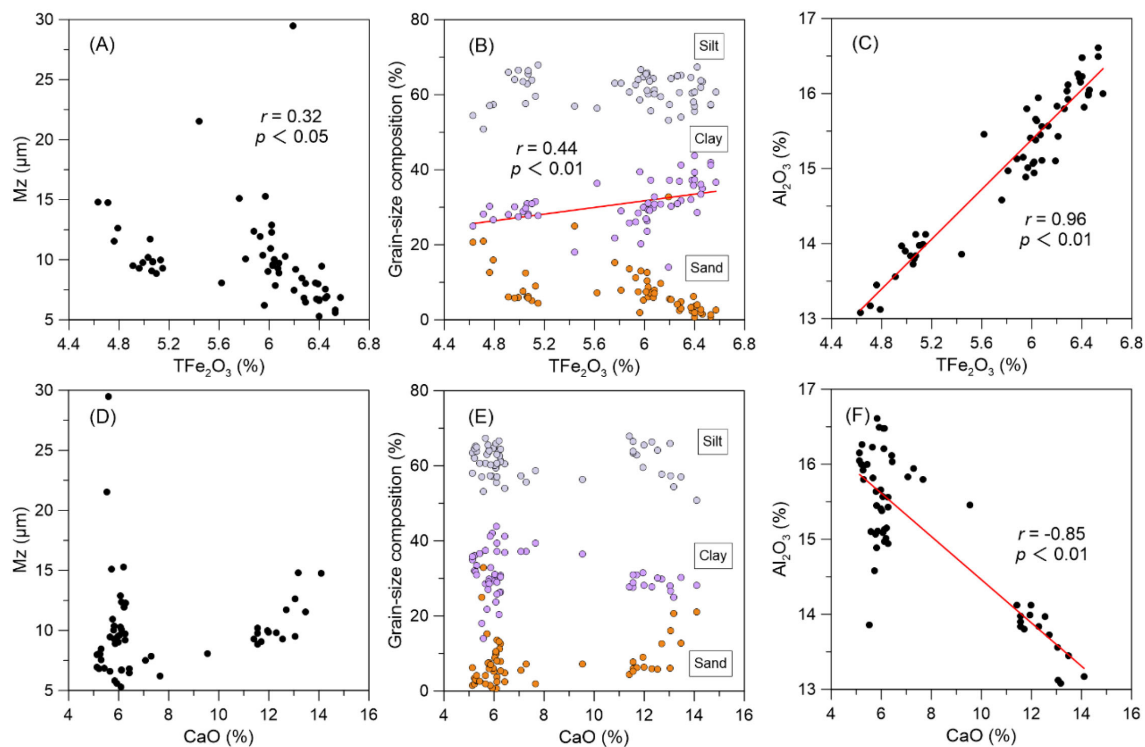


FIGURE 5

Relationships between sedimentary parameters and geochemical compositions in core C1624. (A) Mean grain size (Mz; μm) with TFe_2O_3 content (%). (B) Grain-size composition (%) with TFe_2O_3 content (%), (C) Al_2O_3 content (%) with TFe_2O_3 content (%), (D) Mz (μm) with CaO content (%), (E) Grain-size composition (%) with CaO content (%), (F) Al_2O_3 content (%) with CaO content (%). Data for Mz and grain-size composition are from Wang et al. (2025). Red lines indicate linear regression fits.

5.3 Sea-level control on terrestrial input since the LGM

Given that much of the modern ECS shelf is shallower than 120 m, the distance between the OT and the coastline was highly sensitive to sea-level changes during the deglaciation. Consequently, sea-level changes since the LGM have likely exerted a significant influence on the delivery of terrestrial materials to the OT (Chen et al., 2023; Diekmann et al., 2008). To gain a deeper understanding of the macro-scale mechanisms controlling variations in terrestrial and biogenic input, we placed the terrestrial input index ($\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$) and the biogenic contribution index ($\text{CaO}/\text{Al}_2\text{O}_3$) from core C1624 within the framework of regional paleoenvironmental evolution for comparative analysis (Figure 6). The results reveal that since the LGM, sediment provenance and depositional processes in the study area exhibited a clear, stage-wise response to sea-level changes (Stages S1–S3).

Stage 1 (25.4–11.6 ka BP): This stage corresponds to the LGM and early deglaciation, characterized by low sea levels (Figure 6E). During the LGM, global sea level was 120–130 m lower than present (Kao et al., 2006b; Lambeck and Chappell, 2001). This caused the coastline to prograde eastward, exposing almost the entire Yellow Sea and most of the ECS continental shelf (Qin et al., 1996; Saito et al., 1998). High-resolution seismic profiles indicate that paleo-Changjiang channel systems were widely distributed on the exposed ECS shelf during the LGM and the deglaciation (Li et al., 2005; Xu et al., 2022), implying that lower sea level allowed the paleo-Changjiang river system to extend across the exposed shelf, providing a direct conduit for

sediment delivery. Concurrently, the KC was weak or absent during this period, which also favored the transport and deposition of fluvial sediments (Li et al., 2018; Ujiie et al., 2003; Zheng et al., 2014). The $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratio remained consistently high throughout this period (Figure 6B). Similarly, the $\text{Fe}_2\text{O}_3/\text{TiO}_2$ ratio in core C01 from the western slope of the middle OT (location in Figure 1) also showed high values during this stage (Xu et al., 2025). In contrast, the East Asian summer monsoon (EASM) was weaker during the glacial and deglacial periods compared to the Holocene (Figure 6A) (North Greenland Ice Core Project Members, 2004), suggesting relatively lower riverine input, which stands in opposition to the observed high $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratios. Furthermore, the EASM exhibited strong fluctuations during the deglaciation (Figure 6A; Dykoski et al., 2005), but this signal is not reflected in the $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ record of core C1624, the $\text{Fe}_2\text{O}_3/\text{TiO}_2$ record of core C01, or the Sr-Nd isotopic record of core DGKS9604 (Figures 6A–C; Dou et al., 2012; Xu et al., 2025). This indicates that terrestrial input to the OT during the glacial and deglacial periods was primarily governed by sea level. Based on previous reconstructions of the paleo-channels of the Huanghe and Changjiang (Li et al., 2005; Liu et al., 2010; Ujiie and Ujiie, 1999; Xu et al., 2022), core C1624 is located closer to the paleo-Changjiang system. Sr-Nd isotopic records from the nearby core DGKS9604 on the western slope of the middle OT (Figure 6C, core location in Figure 1; Dou et al., 2012) further support the interpretation that sediments deposited during this period were likely predominantly sourced from the Changjiang. The lower $\text{CaO}/\text{Al}_2\text{O}_3$ ratio (Figure 6D) suggests that biogenic carbonate production and/or preservation was

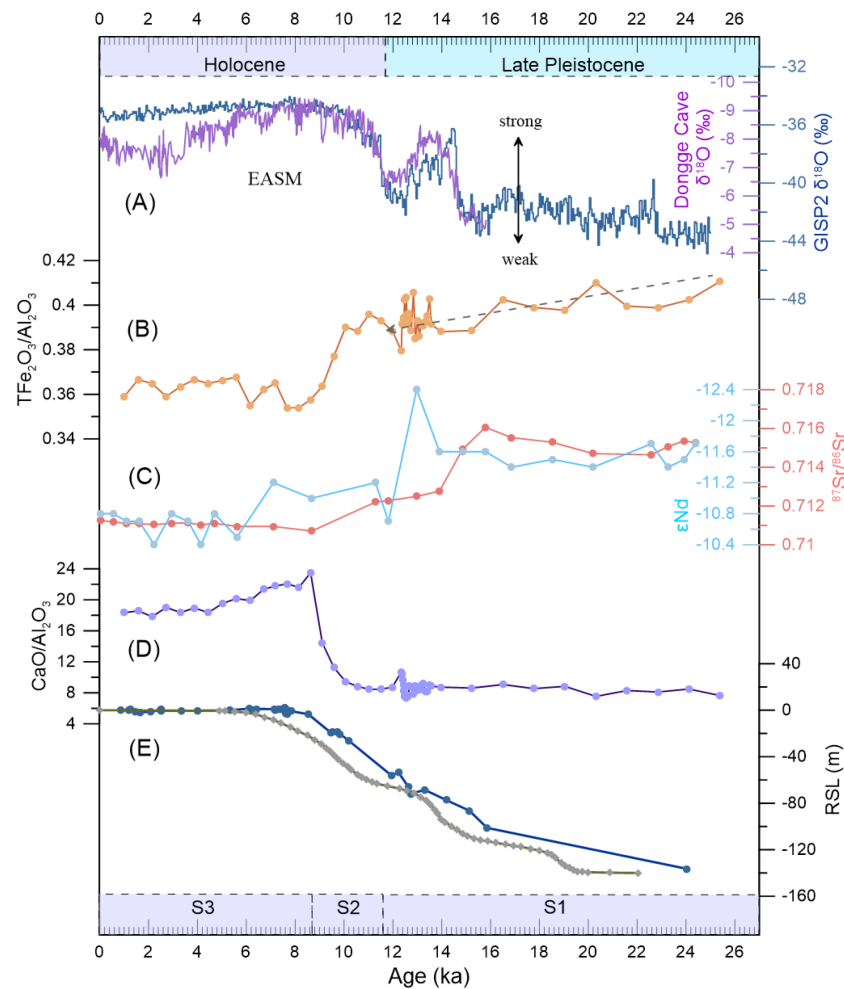


FIGURE 6

Response of terrestrial input to sea-level and climate changes since the Last Glacial Maximum. (A) EASM reconstructions with different indicators. The blue line represents the NGRIP $\delta^{18}\text{O}$ record (North Greenland Ice Core Project Members, 2004). The purple line represents $\delta^{18}\text{O}$ record in Dongge Cave, China (Dykoski et al., 2005). (B) $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratio of core C1624. (C) ϵNd values (blue line) and $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (pink line) of core DGKS9604 (Dou et al., 2012). (D) $\text{CaO}/\text{Al}_2\text{O}_3$ ratio of core C1624. (E) Relative sea level. The blue line is from Lambeck et al. (2014). The black curve is from Li et al. (2014).

suppressed, likely due to enhanced terrestrial detrital input (dilution effect) and/or relatively cold glacial marine conditions.

Stage 2 (11.6–8.7 ka BP): This stage corresponds to a period of rapid global sea-level rise (Figure 6E) (Lambeck et al., 2014; Li et al., 2014). The $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratio decreased sharply from its high values at the end of S1 (Figure 6B), marking the rapid attenuation of the S1 source input and instability in the transport process. Sediment discharged from the Changjiang began forming its modern delta at this time (Li et al., 2000, 2002). Mineralogical characteristics of core C1624 indicate a mixed provenance from the Changjiang and western Taiwan rivers during the early Holocene, with decreasing contribution from the Changjiang and increasing influence from western Taiwan rivers (Wang et al., 2025). The Sr–Nd isotopic data from core DGKS9604 began a fundamental shift: ϵNd values started a significant positive excursion, accompanied by a synchronous decrease in $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (Figure 6C), indicating a markedly increased contribution of materials from Taiwan (Dou et al., 2012). This shift is attributed to profound changes in sediment dispersal patterns, driven by rapid sea-level rise. The inundation of the

previously exposed continental shelf displaced the Changjiang estuary farther from the study area. Concurrently, the KC strengthened and/or re-entered the OT in the later phase of the deglaciation, acting as an effective agent for transporting Taiwanese materials northward (Dou et al., 2010b; Wang et al., 2024). These dramatic changes in paleogeography and circulation patterns profoundly reconfigured sediment transport pathways. The $\text{CaO}/\text{Al}_2\text{O}_3$ ratio showed a rapid increasing trend during this stage (Figure 6D). This may reflect the recovery of marine productivity under a warming climate (Figure 6A) (Dykoski et al., 2005; North Greenland Ice Core Project Members, 2004), and is also attributable to the sharp decrease in terrestrial detrital input flux, which reduced the dilution of biogenic components.

Stage 3 (8.7–0 ka BP): Entering the mid Holocene, sea level stabilized at a high level (Figure 6E) (Lambeck et al., 2014; Li et al., 2014), and the modern circulation pattern became established (Kim and Kucera, 2000; Li et al., 2009), with the main axis of the KC stably flowing through the OT (Kao et al., 2006b; Lee et al., 2013). Under this circulation regime, sediments from the Changjiang are largely

deposited near the estuary or transported southwestward to form the mud belt of the ECS inner shelf (Bian et al., 2013; Liu et al., 2007b; Zhang et al., 2023). Conversely, sediments from western Taiwan can reach the OT region via the Taiwan warm current (Dou et al., 2016; Hsiung and Saito, 2017; Wang et al., 2024). Studies of drifter trajectories in the Taiwan Strait and ocean circulation models around Taiwan show that sediments from western Taiwan flow northeastward through the Taiwan Strait, then enter the area north of Taiwan, eventually reaching the southern and middle OT via the main Kuroshio stream (Jan et al., 2002; Tseng and Shen, 2003). The $\text{TFe}_2\text{O}_3/\text{Al}_2\text{O}_3$ ratio remained generally low during S3 (Figure 6B). The $\text{Fe}_2\text{O}_3/\text{TiO}_2$ ratio in core C01 was also in a low-value range during this stage compared to the last glacial and deglacial periods (Xu et al., 2025). The stable Sr-Nd isotopic signature observed in core DGKS9604, with $^{87}\text{Sr}/^{86}\text{Sr}$ ratios confined to a narrow range (0.7107–0.7112) and ϵNd values remaining high (averaging -10.7), clearly indicates an established and predominant contribution from Taiwanese sources (Dou et al., 2012).

In summary, the record from core C1624 reveals a multi-stage control pattern on sediment input in the southern middle OT since the LGM. On glacial-interglacial scales, sea-level change emerges as the paramount factor reshaping the regional depositional pattern and driving provenance transitions.

6 Conclusions

The OT preserves continuous and complete sedimentary records since the Quaternary, containing abundant paleoclimatic and environmental information on sea-level fluctuations, sediment source-to-sink processes, and KC evolution, making it a key region for studying East Asian paleoenvironmental evolution and sediment dispersal systems along continental margins. Based on comprehensive analysis of data from core C1624 in the southern part of the middle OT, this study reveals that its sediments are primarily composed of a mixture of two end-members: terrigenous detritus and marine biogenic carbonate. Since the LGM, terrestrial input has exhibited a clear three-stage evolutionary pattern. On glacial-interglacial scales, sea-level change emerges as the paramount factor reshaping the regional depositional pattern and driving the provenance transition.

Based on core C1624 from the western slope of the middle OT, and in conjunction with previously published data from core C01 (Xu et al., 2025) and core DGKS9604 (Dou et al., 2012) nearby, this study has preliminarily elucidated the controlling role of sea-level change on terrestrial input in the middle OT. However, several issues require further investigation. Future work could proceed in the following directions: 1) Combining higher-resolution geochemical indicators (e.g., Sr-Nd-Pb isotopes and mineral assemblages) to quantitatively resolve the mixing proportions of materials derived from the Changjiang, Huanghe, and Taiwan during different periods; 2) Collecting more cores from the southern, northern, and eastern slopes of the OT to conduct multi-proxy spatial comparative studies, verifying the regional consistency of the provenance evolution pattern described above

and refining the characterization of transport pathways for materials from different source regions; 3) Although sea level influences provenance by altering circulation, the specific physical processes involved (e.g., precise variations in KC intensity and axis position) require further constraint and quantification using more direct paleoceanographic circulation proxies to establish a more precise “sea level–circulation–provenance” linkage response model.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material. Further inquiries can be directed to the corresponding authors.

Author contributions

MZ: Formal Analysis, Supervision, Writing – original draft, Visualization, Writing – review & editing. LW: Writing – review & editing, Supervision, Data curation, Resources. JZ: Project administration, Writing – review & editing. YZ: Funding acquisition, Writing – review & editing. HT: Writing – review & editing. SC: Writing – review & editing. BH: Writing – review & editing. JK: Writing – review & editing. FW: Writing – review & editing. XL: Writing – review & editing, Writing – original draft, Conceptualization.

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Conflict of interest

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Supplementary material

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